Lab 2.4

Propagate a Default Route in OSPFv2

Default Route Propagation

Default Route Propagation

Propagate a Default Static Route in OSPFv2

To propagate a default route, the edge router must be configured with the following:

- A default static route using the **ip route 0.0.0.0 0.0.0.0** [*next-hop-address* | *exit-intf*] command.
- The **default-information originate** router configuration command. This instructs R2 to be the source of the default route information and propagate the default static route in OSPF updates.

In the example, R2 is configured with a loopback to simulate a connection to the internet. A default route is configured and propagated to all other OSPF routers in the routing domain.

Note: When configuring static routes, best practice is to use the next-hop IP address. However, when simulating a connection to the internet, there is no next-hop IP address. Therefore, we use the *exit-intf* argument.

```
R2(config)# interface lo1
R2(config-if)# ip address 64.100.0.1 255.255.255.252
R2(config-if)# exit
R2(config)# ip route 0.0.0.0 0.0.0.0 loopback 1
%Default route without gateway, if not a point-to-point interface, may impact performance
R2(config)# router ospf 10
R2(config-router)# default-information originate
R2(config-router)# end
```

Verify the Propagated Default Route

- You can verify the default route settings on R2 using the **show ip route** command. You can also verify that R1 and R3 received a default route.
- Notice that the route source on R1 is **O*E2**, signifying that it was learned using OSPFv2. The asterisk identifies this as a good candidate for the default route. The E2 designation identifies that it is an external route.

```
R2# show ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
S*        0.0.0.0/0 is directly connected, Loopback1
      10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
(output omitted)
```

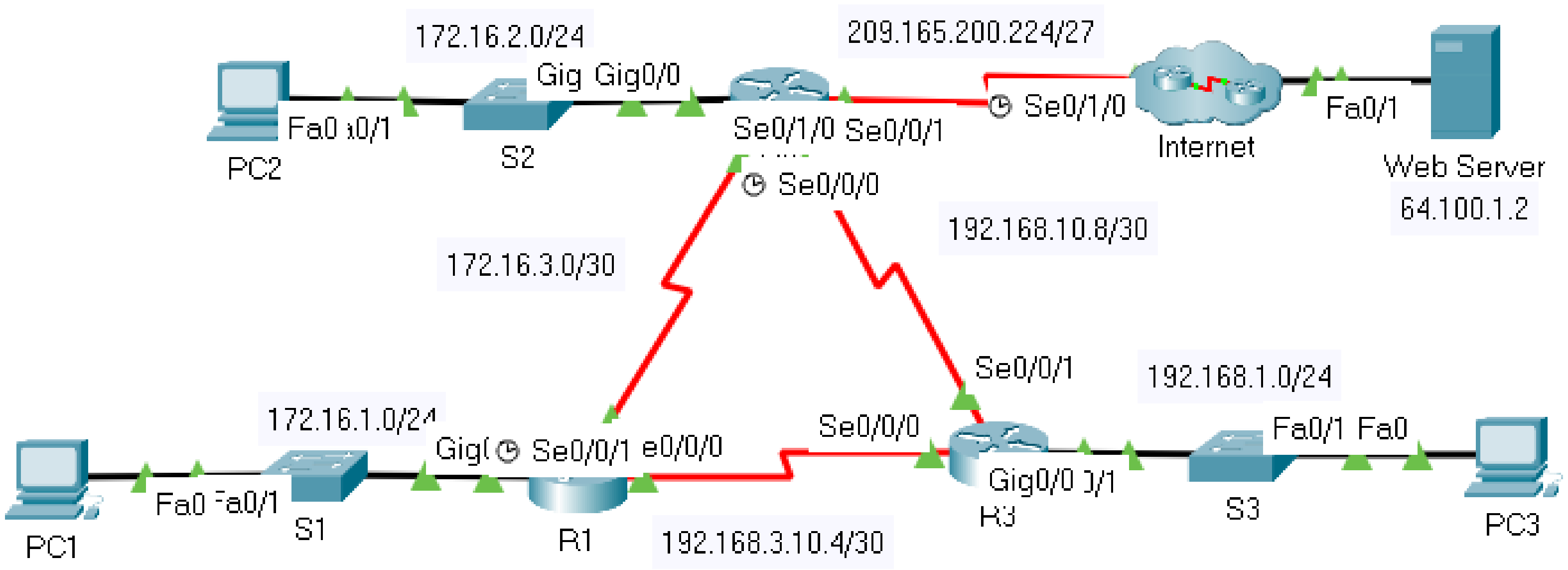
```
R1# show ip route | begin Gateway
Gateway of last resort is 10.1.1.6 to network 0.0.0.0
O*E2  0.0.0.0/0 [110/1] via 10.1.1.6, 00:11:08,
GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
(output omitted)
```

Packet Tracer - Propagate a Default Route in OSPFv2

In this Packet Tracer activity, you will complete the following:

- Propagate a Default Route
- Verify Connectivity

Lab 2.4 Propagate a Default Route in OSPFv2



Lab 2.4 Propagate a Default Route in OSPFv2

1. Configure a default route on R2.

Configure R2 with a directly attached default route to the internet.

R2(config)# ip route 0.0.0.0 0.0.0.0 Serial0/1/0

2. Propagate the route in OSPF.

Configure OSPF to propagate the default route in OSPF routing updates.

R2(config)# router ospf 1

R2(config-router)# default-information originate

Lab 2.4 Propagate a Default Route in OSPFv2

3. Examine the routing tables on R1 and R3.

Examine the routing tables of R1 and R3 to verify that the route has been propagated.

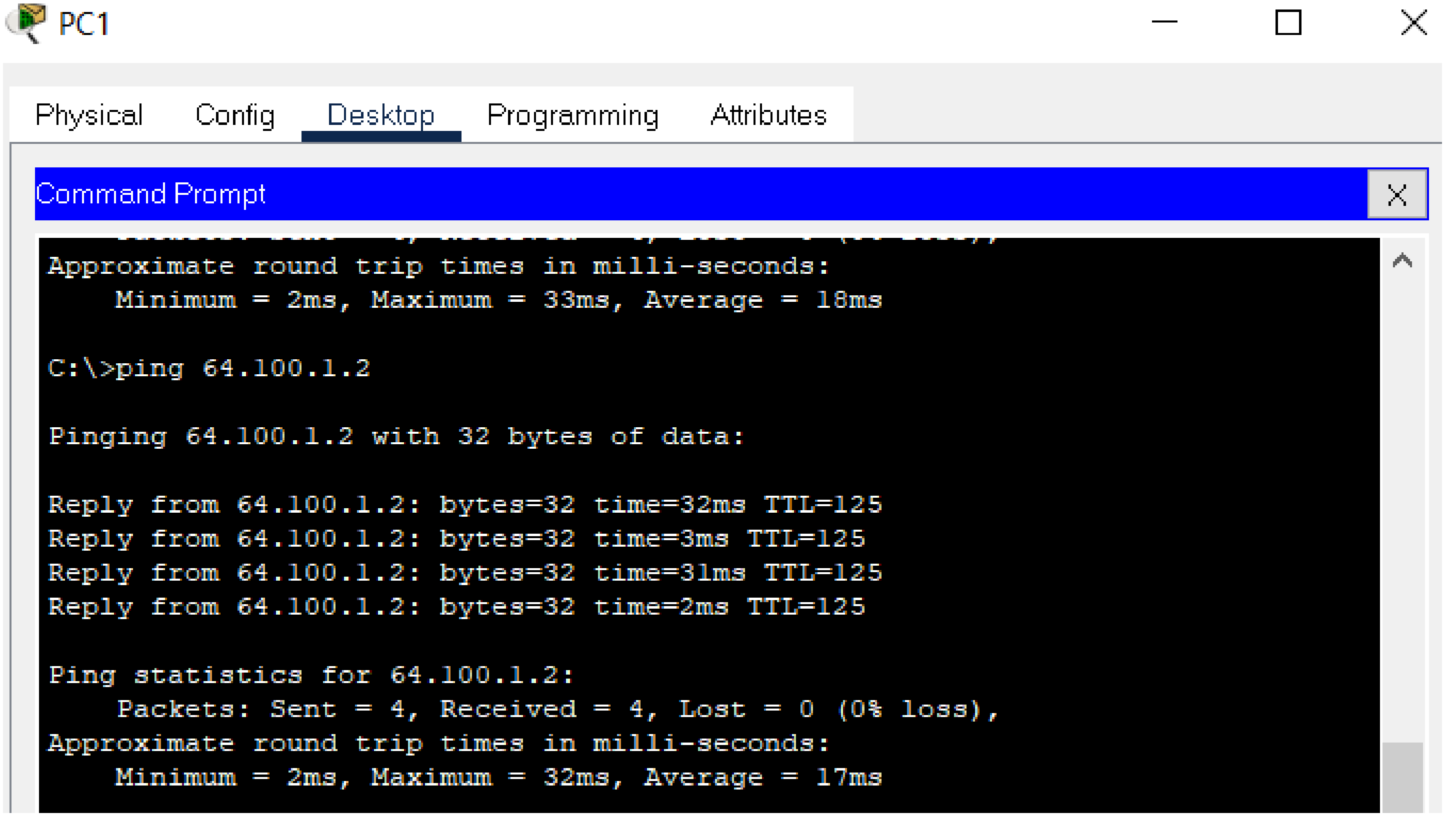
```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.3.2 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:22:53, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.6, 00:22:53, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 192.168.10.6, 00:22:53, Serial0/0/1
                        [110/128] via 172.16.3.2, 00:22:53, Serial0/0/0
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 172.16.3.2, 00:22:53, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:12:49, Serial0/0/0
```

```
R3>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O       172.16.1.0/24 [110/65] via 192.168.10.5, 00:23:53, Serial0/0/0
O       172.16.2.0/24 [110/65] via 192.168.10.9, 00:23:53, Serial0/0/1
O       172.16.3.0/30 [110/128] via 192.168.10.9, 00:23:53, Serial0/0/1
                      [110/128] via 192.168.10.5, 00:23:53, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/0
L       192.168.10.6/32 is directly connected, Serial0/0/0
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.10/32 is directly connected, Serial0/0/1
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 192.168.10.9, 00:23:53, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:13:54, Serial0/0/1
```

Lab 2.4 Propagate a Default Route in OSPFv2

4. Verify Connectivity

Verify that PC1, PC2, and PC3 can ping the web server.



Lab 2.4 Propagate a Default Route in OSPFv2

4. Verify Connectivity

Verify that PC1, PC2, and PC3 can ping the web server.

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

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